

Kanisha Arora

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Summary

Computer Science & Engineering student with hands-on experience in Deep Reinforcement Learning research, cybersecurity simulation, data pipelines, full-stack development, and embedded IoT systems. Skilled in Python, SQL, React, Next.js, Streamlit, REST APIs, anomaly detection, ESP32, BLE, and simulation-based engineering workflows. Strongest work includes federally funded research, energy-system cybersecurity research, and an award-winning senior design project.

Technical Skills

Programming: Python, Java, C, C++, SQL, JavaScript, HTML, CSS

Web, Data & Tools: React, Next.js, Node.js, Streamlit, Pandas, REST APIs, Firebase, Clerk, Git, VS Code, IntelliJ, MATLAB

Research & Systems: Deep Reinforcement Learning, simulation workflows, anomaly detection, PCAP/DNS analysis, digital twins, Linux, Windows, macOS, ESP32, BLE, Wi-Fi video streaming

Education

University of Toledo, Ohio

Expected Aug 2026

Bachelor of Science in Computer Science & Engineering

GPA: 3.7

Experience

Research Co-op - Deep Reinforcement Learning for Electric Ships

May 2026 - Present

University of Toledo, Department of Electrical Engineering and Computer Science

- Contribute to a federally funded project developing a Deep Reinforcement Learning framework for Electric Ship energy and control systems.
- Study, run, and modify DRL simulation workflows for training, testing, model evaluation, reward trends, and system-response analysis.
- Support INL-connected electric ship research by reviewing control behavior, evaluation outputs, implementation progress, and technical challenges with faculty and graduate mentors.

Research - Cybersecurity of Smart Grids

May 2025 - Aug 2025

University of Toledo, Department of Electrical Engineering and Computer Science

- Investigated cybersecurity vulnerabilities in hydrogen-based Integrated Energy Systems using simulation and digital twin models.
- Designed cyber-attack scenarios and anomaly/intrusion-detection logic to evaluate robustness and identify abnormal system behavior.
- Analyzed system stability under simulated attacks, proposed mitigation strategies, and documented findings in a technical report and presentation.

Technical Projects

RC Trailer Jack - 1st Place, 2026 KEEN Senior Design Capstone Project Competition

Spring 2026

- Designed and built a smartphone-controlled motorized trailer jack to automate trailer parking and hitching in tight spaces.
- Integrated mechanical hardware, embedded firmware, motor control, Android app logic, ESP32 communication, BLE commands, and Wi-Fi video streaming into one working system.
- Developed real-time controls for throttle, steering, braking, gear selection, emergency stop, command timeout, and automatic motor shutdown on disconnect.
- Validated system performance through iterative prototype and full-scale trailer testing under loaded and unloaded conditions.

DNS Tunneling Detection System

Spring 2026

- Built a Python tool to analyze PCAP traffic and flag suspicious hosts using query volume, unique domains, TXT usage, query length, and entropy-based randomness indicators.
- Implemented configurable anomaly scoring, alert outputs, and visual graphs to support network security investigation and suspicious traffic review.

Job Market Analytics Dashboard

Spring 2026

- Built a Python/Streamlit dashboard integrating Greenhouse and Lever APIs to process 1,900+ U.S.-based postings with cleaned JSON data, state filters, search, and 7-day recency filtering.
- Developed a Pandas ETL workflow to normalize inconsistent API data and improve job-search usability through structured filtering and dashboard views.

Additional Experience & Projects

- **Kids Camp Counselor, UToledo Recreational Services** (*May 2026 - Present*): Support camper supervision, daily activities, swim-day monitoring, safety procedures, communication, and team-based recreation operations.
- **Academic Industry Immersion Seminar - Database & ETL Project** (*May 2024 - Aug 2024*): Completed for 1 co-op credit; designed an SQL database, built a Python ETL workflow, and presented technical outcomes.
- **Invest-In-Box / Tech Market:** Built software projects involving Python, SQL, API integration, Firebase, Clerk authentication, full-stack development, and AI-powered profile evaluation using Llama/OpenAI models.